

LONDON'S GLOBAL UNIVERSITY

UCL Corporate Policy and Regulation:

Electrical Safety, Testing, and Maintenance

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Contents

1	Introduction, Regulatory Framework, and Scope	3
1.1	Introduction	3
1.2	The Electricity at Work Regulations 1989 (EWR)	3
1.3	BS 7671: The IET Wiring Regulations	4
1.3.1	Scope and Application	4
1.3.2	The Principle of Automatic Disconnection of Supply (ADS)	4
1.4	The Provision and Use of Work Equipment Regulations 1998 (PUWER)	4
1.5	The Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR)	5
1.5.1	Identification of Dangerous Substances and Atmospheres	5
1.5.2	ATEX Zoning and Equipment Selection	6
1.5.3	Mitigation, Extraction, and Containment	6
1.5.4	Information and Emergency Procedures	6
1.6	The Supply of Machinery (Safety) Regulations 2008	7
1.6.1	Integration of Electrical and Mechanical Safety	7
1.6.2	Guarding and Safety Interlocks	7
1.6.3	Safety-Critical Control Systems	7
1.6.4	Emergency Stop (E-Stop) Architecture	8
1.6.5	Documentation and Design Justification	8
1.7	Manual Scope and Equipment Coverage	8
1.8	Policy Philosophy: Empowering Advanced Research	8
2	The ESPeR Competency Framework and Operational Autonomy	10
2.1	The Philosophy of Competency at UCL	10
2.2	Top-Down Management and Principal Investigator (PI) Accountability . .	10
2.3	Tiers of Competency and Authorization	11
2.3.1	Standard Operative (CAT I)	11
2.3.2	Advanced Operative (CAT II)	11
2.3.3	Specialist Operative (CAT III)	12
2.4	Equipment Category Definitions	12
2.5	Simplified Permit to Energise (PTE)	12
2.6	Advancement and CPD	12
3	Technical Testing and Maintenance Protocols	13
3.1	Maintenance Intervals and Testing Frequencies	13
3.2	Mandatory Electrical Testing Procedures (FAT & PAT)	13
3.3	Earthing and Equipotential Bonding Thresholds	14
3.4	Key Technical Formulas	14

4	Administration and Information Management	15
4.1	Centralised Information Repository	15
4.1.1	The Electrical Competence App (Front-End Access)	15
4.1.2	Centralised Database (Back-End Administration)	15
4.2	The "Permit to Energize" System	16
4.2.1	Operational Workflow:	16
4.3	Competency Maintenance and CPD	16
5	Hazard Mitigation and Emergency Procedures	17
5.1	Primary Hazards and Mitigation Strategies	17
5.2	Safety Interlocks and Active Monitoring	17
5.3	Emergency Procedures	18
5.4	The "Two-Person Rule"	18

Chapter 1

Introduction, Regulatory Framework, and Scope

1.1 Introduction

University College London (UCL) is committed to providing a safe environment for advanced research, innovation, and learning. Across its various departments and campuses, the university houses an extensive and diverse array of electrical equipment, ranging from standard commercial hardware to bespoke, high-energy experimental systems. These systems vary significantly in supply, ranging from 13A single-phase to 160A three-phase installations, including fixed, mobile, and remote setups.

This manual establishes the regulatory and operational standards for all non-infrastructure electrical equipment across the university. It is designed to empower researchers, particularly PhD and master's students, by providing a clear, non-restrictive path to autonomy while ensuring absolute compliance with statutory safety laws.

1.2 The Electricity at Work Regulations 1989 (EWR)

The **Electricity at Work Regulations 1989** (EWR) is the primary statutory instrument governing electrical safety in the United Kingdom. All personnel at UCL, regardless of their seniority or academic status, must adhere to these regulations.

The EWR requires that:

- All electrical systems are constructed and maintained to prevent danger.
- Work is carried out by competent persons or under such supervision as is appropriate for the nature of the work.
- Equipment is regularly inspected and tested to ensure continued safety.

UCL recognizes that as an employer, the university has a legal duty to be as competent as the individuals performing the work. This is achieved through the maintenance of a competency register, training logs, and robust safety policies.

1.3 BS 7671: The IET Wiring Regulations

While the EWR provides the legal framework, **BS 7671** (The IET Wiring Regulations) provides the technical requirements for achieving safety.

1.3.1 Scope and Application

At UCL, BS 7671 is the baseline for conformity. It ensures that users are protected from electric shock and fire through standardised testing and installation practices. For specialised experimental work where the scope of BS 7671 may not suffice, supplemental standards such as IPC/WHMA-A-620E (for cable and harness assembly) or UN R100 (for EV approval) are applied.

1.3.2 The Principle of Automatic Disconnection of Supply (ADS)

A core requirement of BS 7671 used throughout this manual is the achievement of **Automatic Disconnection of Supply (ADS)**. ADS ensures that in the event of a fault, the protective device (such as an MCB or RCD) disconnects the supply fast enough to prevent lethal electric shock. This is achieved through:

- **Earthing:** Providing a Circuit Protective Conductor (CPC) as a path for fault current.
- **Equipotential Bonding:** Connecting metalwork (pipes, enclosures) to the earth point to ensure all surfaces remain at the same electrical potential during a fault, preventing "touch voltages".

1.4 The Provision and Use of Work Equipment Regulations 1998 (PUWER)

While the EWR governs the electrical danger of a system, PUWER governs the overall safety, suitability, and mechanical integrity of the equipment itself. Within the context of UCL laboratories, "work equipment" encompasses everything from off-the-shelf soldering irons to bespoke, researcher-built high-voltage battery testing rigs and rotating machinery.

Under PUWER, UCL and its Principal Investigators (PIs) have a strict legal duty to ensure that all equipment provided for use is:

- **Suitable for the Intended Use (Regulation 4):** Equipment must be constructed or adapted to be suitable for the specific purpose and conditions in which it will be used. For Category II experimental works, this means the design must account for foreseeable environmental stresses, such as continuous high-current loads, vibration from motors, or potential chemical exposure. Equipment designed for dry office use cannot simply be repurposed for a wet lab without a formal re-evaluation of its suitability.
- **Maintained in a Safe Condition (Regulations 5 & 6):** PUWER legally mandates that equipment must be maintained in an efficient state, in efficient working order, and in good repair.

1. **Preventative Maintenance:** This regulation provides the legal backing for the mandatory testing intervals (e.g., PAT, dead testing, and Z_s verification) detailed in Chapter 3 of this policy.
 2. **Inspection Logs:** Researchers must document maintenance and modifications. Failure to maintain an up-to-date modification log for Category II equipment is a direct breach of this regulation.
- **Information, Instructions, and Training (Regulations 8 & 9):** No individual is permitted to use work equipment unless they have received adequate training and have access to clear operating instructions.
 1. **Information Accessibility:** Risk Assessments (RAs) and Standard Operating Procedures (SOPs) must be immediately available to the user. This is facilitated by the digital centralised repository (e.g., via QR codes on the equipment linking to the Electrical Competence App).
 2. **Competency (ESPER):** Training must be proportional to the risk. The ES-PER tier system (Standard, Advanced, Specialist) directly satisfies the POWER requirement that only trained personnel may operate complex or high-risk equipment.
 - **Controls, Isolation, and Emergency Stops (Regulations 14–19):** Experimental rigs often integrate multiple power sources and mechanical hazards. POWER strictly requires that
 1. **Isolation:** All equipment must have a clearly identifiable and accessible means of isolating it from all energy sources (lock-out/tag-out capability).
 2. **Emergency Power Off (EPO):** Systems must be equipped with readily accessible emergency stop controls unless an RA demonstrates it would not lessen the risk.
 3. **Control Systems:** The start/stop controls must be safe. A system must never be able to restart automatically after a power failure or emergency stop reset if doing so poses a danger.

1.5 The Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR)

DSEAR places a statutory duty on employers to protect personnel from the risks of fire, explosions, and similar events arising from dangerous substances used or present in the workplace. In the context of UCL laboratories, electrical equipment is a primary potential ignition source. The regulations mandate that electrical systems must not introduce an ignition risk into an explosive atmosphere.

1.5.1 Identification of Dangerous Substances and Atmospheres

Before any Category II experimental rig is designed or energized, researchers must assess whether dangerous substances will be present. For electrical and propulsion research, these commonly include:

- **Battery Off-Gassing:** Lithium-ion thermal runaway or failure events release highly flammable and toxic gases (e.g., hydrogen, carbon monoxide, and vapourised electrolyte).
- **Flammable Gases:** Hydrogen utilised in fuel cell test beds or specialised gaseous testing environments.
- **Volatile Solvents:** Chemicals used in cleanroom microfabrication, lithography processes, or general wet-lab areas.

1.5.2 ATEX Zoning and Equipment Selection

Where an explosive atmosphere may occur, the area must be formally classified into hazardous zones (Zone 0, 1, or 2 for gases/vapours) based on the frequency and duration of the hazard.

- **Ex-Rated Equipment:** Standard electrical enclosures and glands (e.g., standard IP68 nylon glands) are strictly prohibited in zoned areas. All electrical equipment placed within a hazardous zone must be ATEX-certified (e.g., Ex d for flameproof enclosures or Ex e for increased safety).
- **Intrinsically Safe Design:** Where possible, sensing and control circuits entering a hazardous zone must be designed as intrinsically safe (Ex i), utilising Zener barriers or galvanic isolators to ensure the electrical energy available is insufficient to cause ignition.

1.5.3 Mitigation, Extraction, and Containment

DSEAR requires that risks be eliminated or reduced as far as is reasonably practicable. For electrical testing, this requires engineering controls:

- **Active Extraction:** High-energy battery testing must be conducted under active extraction systems (e.g., fume hoods or snorkel exhausts) interlocked with the power supply. If extraction fails, the electrical test must automatically abort.
- **Explosion-Proof Enclosures:** Where destructive testing or thermal runaway is a foreseeable risk, batteries and relevant power electronics must be housed within blast-rated containment boxes capable of withstanding the overpressure of an explosive event.

1.5.4 Information and Emergency Procedures

Personnel operating in or near DSEAR zones must be specifically briefed on the hazards. The Lab RA must explicitly detail the emergency response to a localised fire or gas release, specifically prohibiting the opening of containment chambers until stabilisation is achieved and vapours have dissipated.

1.6 The Supply of Machinery (Safety) Regulations 2008

While these regulations primarily govern the commercial sale of machinery, they apply equally to bespoke machinery designed and constructed in-house for use by university staff and students.

The Electrical Competence Group requires that all Category II experimental machinery meets the Essential Health and Safety Requirements (EHSRs) outlined in these regulations before a Permit to Energise can be issued.

1.6.1 Integration of Electrical and Mechanical Safety

Electrical faults must not result in mechanical hazards. The design of the machinery must ensure that:

- Loss of power, voltage fluctuations, or complete electrical failure does not lead to a dangerous situation (e.g., dropped loads, runaway motors, or failure of cooling systems).
- Re-establishment of power after an outage does not cause an unexpected or spontaneous restart of the machinery.

1.6.2 Guarding and Safety Interlocks

Where physical hazards (such as high-speed rotating shafts, drive belts, or automated actuators) exist, physical guarding is required.

- **Interlocked Guards:** Fixed or movable guards must be integrated with the electrical control system. Opening a guard must immediately break the control circuit (via mechanically linked safety switches or RFID interlocks), safely halting all moving parts before the operator can reach the hazard zone.
- **Defeatability:** Safety interlocks must be designed so they cannot be easily bypassed, defeated, or overridden by the operator.

1.6.3 Safety-Critical Control Systems

The reliability of the control system is critical. The electrical design must incorporate "fail-safe" principles.

- **Fault Tolerance:** A fault in the hardware or software of the control system must not lead to hazardous situations. For example, a short circuit in a sensor cable must not cause a motor controller to command maximum torque.
- **Redundancy:** For high-risk machinery, dual-channel safety relays and redundant contactors should be utilized so that the failure of a single electrical component does not compromise the safety function.

1.6.4 Emergency Stop (E-Stop) Architecture

Every machine must be fitted with one or more emergency stop devices to enable actual or impending danger to be averted.

- **Override Priority:** The E-Stop command must override all other operating commands.
- **Hardware Intervention:** The E-Stop must operate via a hardwired electromechanical circuit that physically disconnects power to the machine actuators (e.g., dropping out a main contactor). Relying solely on software (e.g., a software "stop" command to a PLC or microcontroller) is strictly prohibited for primary emergency stops.

1.6.5 Documentation and Design Justification

For all bespoke machinery, Advanced Operatives (CAT II) must document their safety design within the risk assessment. This includes detailing how moving parts are guarded, providing a schematic of the interlock and E-Stop circuits, and proving that the machine operates safely under both normal conditions and single-fault conditions.

1.7 Manual Scope and Equipment Coverage

This manual covers all non-infrastructure electrical equipment across UCL laboratories and facilities. This includes:

- **Standard Equipment:** Computers, laptops, soldering stations, and analytical equipment.
- **Power Systems:** Power supplies, electric motors, and battery cyclers.
- **Safety and Containment:** Battery safety enclosures and fume hoods.
- **Custom Research Systems:** Bespoke cabling, customized power electronic interfaces, and experimental test beds.

1.8 Policy Philosophy: Empowering Advanced Research

This manual rejects a one-size-fits-all approach to safety. It recognizes that PhD and Master's students in Electrical Engineering possess advanced theoretical knowledge that must be balanced with practical lab-specific competency.

The policy allows for:

- **Self-Authorization:** Competent researchers can sign off on their own "Permit to Energise" for routine Category I and II tasks once initial baseline safety is established.
- **Administrative Support:** A dedicated SharePoint infrastructure provides easy access to maintenance logs, RAs, and work permits, reducing the administrative burden on researchers.

- **Scalable Safety:** Oversight increases only when the risk increases (e.g., high-voltage pack assembly vs. single-cell testing), ensuring that specialised research is not delayed.

Chapter 2

The ESPeR Competency Framework and Operational Autonomy

Chapter 2 outlines the administrative and technical structure of the UCL Electrically Skilled Persons Register (ESPeR). This framework is designed to categorise the university's diverse workforce—from undergraduate students to specialist lead researchers—into clear tiers of authority and responsibility.

2.1 The Philosophy of Competency at UCL

Competency within the university environment is not merely a measure of an individual's academic background; it is a holistic assessment of skill, experience, and the organisation's ability to maintain safety standards. As an employer, UCL has a statutory duty to ensure that its employees are as competent as possible. This is achieved through the implementation of rigorous electrical safety policies and the maintenance of comprehensive training, maintenance, and CPD logs.

2.2 Top-Down Management and Principal Investigator (PI) Accountability

While the ESPER framework empowers researchers with operational autonomy, ultimate accountability relies on robust top-down management. Principal Investigators (PIs) and Lab Managers bear the primary responsibility for the safety and compliance of their respective research groups.

PIs are strictly accountable for ensuring that their undergraduate students, postgraduate researchers, and staff operate exclusively within their authorised competency tier (Standard CAT I, Advanced CAT II, or Specialist CAT III). It is the explicit duty of the PI and lab manager to:

- Verify that no researcher undertakes complex Category II or III work without the documented prerequisite training and formal authorisation.
- Ensure that Risk Assessments (RAs) and Permits to Energise (PTEs) are correctly filed and adhered to for all experimental setups.

- Actively monitor the laboratory environment to prevent "competency creep", where unqualified students attempt modifications beyond their authorised tier.

Failure by management to enforce these tier boundaries constitutes a critical breakdown in institutional competence and safety compliance.

2.3 Tiers of Competency and Authorization

Competency is evaluated through a combination of qualifications, experience, and peer review:

2.3.1 Standard Operative (CAT I)

- **Scope:** Typically undergraduate students, new Master's/PhD researchers, or staff with foundational electrical training.
- **Requirement:** 0-3 years experience or Level 2+ qualification (including specific degree modules).
 - **Mandatory Day 1 Induction:** Regardless of prior academic qualifications, all incoming students and staff must complete a standardised "Basic Electrical Safety" module as part of their initial Day 1 laboratory induction. Physical access to the laboratory (via swipe card or key) will not be granted until this fundamental CAT I competency is documented and verified by the Lab Manager.
- **Authority:** Authorized for the installation and inspection of mobile and fixed equipment within Category I (standard 13A single-phase supplies and single-cell battery systems).
- **Autonomy:** Operates under the supervision or training of an advanced or specialist operative.

2.3.2 Advanced Operative (CAT II)

- **Scope:** Senior PhD researchers, postdoctoral research associates, or experienced technical staff.
- **Requirement:** 3-5 years experience or Level 3+ qualification. For postgraduate researchers, significant academic achievement in electrical engineering is evaluated as equivalent to field experience.
- **Authority:** Authorised for work on equipment up to 63A (single or three-phase) and battery energy storage modules.
- **Autonomy:** Authorised to work alone and to supervise/approve the work of standard operatives.

2.3.3 Specialist Operative (CAT III)

- **Scope:** Lab Managers, Pls, or Specialist Technicians.
- **Requirement:** 5+ years of experience, a Level 4+ qualification, or professional registration with an industry-recognised body (e.g., IET). Required for high-voltage work exceeding 1000 VDC.
- **Authority:** Full authority across Categories I, II, and III, including battery packs and custom load banks; authorised to sign off on Permit-to-Energise forms.
- **Autonomy:** Acts as the final technical authority, with the power to approve work and train all lower tiers.

2.4 Equipment Category Definitions

To ensure clarity in the "Permit to Energise" process, electrical systems are divided into three risk categories based on power and energy levels:

Category	Supply/Power Limit	Battery System Equivalent
Category I	Up to 13A Single-Phase	Single-Cell BESS
Category II	Up to 63A Single/Three-Phase	BESS Modules
Category III	>63A Single/Three-Phase	Full BESS Packs

2.5 Simplified Permit to Energise (PTE)

The Permit to Energise is the primary operational control used before any system is switched on. To support fast-paced research, the PTE system is designed for maximum accessibility:

1. **Self-Issue (CAT I & II):** Once a researcher has reached the Advanced Operative tier, they are empowered to sign off on their own category-compliant setups. They do not require a secondary signature for routine energisation of validated designs.
2. **Digital Submission:** All permits are managed via a centralised SharePoint folder, allowing for rapid recording and auditing without physical paperwork.
3. **Dead Test Prerequisite:** Before any PTE is issued for a new assembly, the operator must document basic "Dead Tests" (e.g., Earth Continuity and Insulation Resistance) in the lab maintenance log.

2.6 Advancement and CPD

UCL encourages a culture of continuous development. Researchers are encouraged to maintain a CPD (Continuing Professional Development) Log to track their progression from Standard to Advanced or Specialist status. Advancement is achieved through a combination of training from higher-tier operatives, formal qualifications, and documented hours of hands-on experience.

Chapter 3

Technical Testing and Maintenance Protocols

Chapter 3 details the specific technical requirements for the inspection, testing, and upkeep of electrical equipment across the university. To ensure the safety of all personnel, the following procedures must be adhered to before and after any installation is energised.

3.1 Maintenance Intervals and Testing Frequencies

Testing and maintenance processes are covered by a specific risk assessment, which must be followed at all times. Testing logs are mandatory for all equipment types. The frequency of inspection is determined by the mobility and supply current of the equipment.

Equipment types and competency required for	New (within 1 year of purchase)		In-use (in use for 1 year or more)	
	Fixed/unlikely to be moved	Mobile/likely to be moved	Fixed/unlikely to be moved	Mobile/likely to be moved
13A supply current Single-phase Up to 230 V Standard competency	No PAT testing required for first year Visual check all cables and connections		PAT test every 3 years If inaccessible/hazardous to access then visual check only, every 6 months	PAT test every year Visual check all cables and connections
>13A – 63 A supply current Single- or three-phase Up to 500 V Advanced competency	Test on receipt/setup Test accessible safety devices (RCDs, RCBOS, earthing, etc.) Visual check all cables and connections	Test on receipt/setup and after 6 months Test accessible safety devices (RCDs, RCBOS, earthing, etc.) Visual check all cables and connections	Test every year Test accessible safety devices (RCDs, RCBOS, earthing, etc.) Visual check all cables and connections	Test every 6 months Test accessible safety devices (RCDs, RCBOS, earthing, etc.) Visual check all cables and connections
>63A supply current Single- or three-phase Any load banks or custom cables Up to 2 kV Specialist competency	Test before each use Test accessible safety devices (RCDs, RCBOS, earthing etc.) Visual check all cables and connections			

3.2 Mandatory Electrical Testing Procedures (FAT & PAT)

Before and after an installation is energised, the following tests must be performed:

- **Dead Tests (De-energised State):** These tests must be conducted before the system is connected to a power source to identify faults without risk of shock: Earth Continuity (essential for confirming the Circuit Protective Conductor (CPC) is intact for fault protection), Insulation Resistance (conducted to ensure conductors are properly isolated. An item with a measured resistance of 1 MΩ or above is considered non-conductive), Flash Test (a high-voltage stress test of the insulation), and Polarity of Conductors (confirms that the line and neutral conductors are correctly identified and connected).
- **Live Tests (Energised State):** Phase Sequence (Verification of the correct order of phases for single and three-phase equipment), Touch Current / Protective Conductor Current (Measuring current leakage to ensure it stays within safe limits), RCD Test (Verifying that the Residual Current Device trips at the correct current and time to prevent lethal shock), Functional Equipment Test (Ensuring the system operates as intended under load).

3.3 Earthing and Equipotential Bonding Thresholds

Protection against electric shock is achieved through Automatic Disconnection of Supply (ADS).

3.4 Key Technical Formulas

- **Conductivity Verification:** Items are non-conductive if resistance is $\geq 1 \text{ M}\Omega$ at 250 V, 500 V, and 1000 V.
- **Bonding Negligibility:** Measured resistance of $\leq 50 \text{ m}\Omega$ between a fitting and earth bond point is electrically negligible.
- **Supplementary Bonding Resistance (R):** In instances where metallic objects require supplementary bonding, the measured resistance (R) must satisfy the following thresholds based on the operating current of the protection device (I_a):
 - **A.C. Circuits:** $R \leq 42.2V/I_a$.
 - **D.C. Circuits:** $R \leq 60V/I_a$.

Chapter 4

Administration and Information Management

Chapter 4 outlines the administrative infrastructure required to support the ESPeR framework and ensure that UCL maintains its institutional competency. To comply with the Electricity at Work Regulations 1989, the university must implement robust record-keeping and distribution systems for all electrical safety data.

4.1 Centralised Information Repository

To comply with the Electricity at Work Regulations 1989 and PUWER 1998, the university must ensure that all skilled persons have immediate access to necessary safety protocols, instructions, and administrative logs. To achieve this and eliminate barriers to information for students, the Electrical Competence Group utilizes a hybrid digital infrastructure.

4.1.1 The Electrical Competence App (Front-End Access)

The Electrical Competence App serves as the primary, immediate interface for students and researchers operating within the laboratories.

- **QR Code Integration:** All active Category II experimental rigs, motor test beds, and high-risk lab benches must be labeled with a unique QR code.
- **Instant Information Delivery:** By scanning the QR code via the app, students and staff will instantly access the specific equipment's active Risk Assessment (RA), Standard Operating Procedure (SOP), the authorised "Permit to Energise", and the date of its last electrical safety test.
- **Emergency Protocols:** The app features a quick-access emergency tab detailing local isolation procedures, the identity of the nearest first aiders, and AED locations.

4.1.2 Centralised Database (Back-End Administration)

A dedicated SharePoint folder (or equivalent LabCup system) acts as the central master database for the working group. This infrastructure supports the App and includes:

- **Administrative Logs:** Training logs, maintenance logs, work permits, and CPD logs.
- **UCL-Specific Documents:** Risk Assessments (RAs) that list relevant parts of each regulatory body.
- **Commercial Standards and Guidance:** Access to essential industry references, including the 18th Edition Wiring Regulations, Guidance Note 3 (Inspection and Testing), and IPC/WHMA-A-620E (Cable and Harness Assembly).

4.2 The "Permit to Energize" System

The Permit to Energise (PTE) is the critical control point between assembly and operation. To ensure this system remains a helpful safety tool rather than an administrative burden, it is designed for rapid execution within the researcher's competency tier.

4.2.1 Operational Workflow:

- **Self-Issue Rights:** Personnel registered as Advanced Operatives or Specialist Operatives are authorized to issue their own permits for work within their respective categories (CAT I & II for Advanced; All CATs for Specialist).
- **Prerequisite Testing:** A permit can only be issued once the mandatory "Dead Tests" (Earth Continuity and Insulation Resistance) have been recorded in the maintenance log.
- **Digital Integration:** Permits are filed digitally via the SharePoint platform, allowing for real-time tracking of active high-energy experiments across the laboratory.

4.3 Competency Maintenance and CPD

Competency is not a static designation; it is a professional status that requires ongoing maintenance. Both the individual and the organisation have a duty to stay current with evolving standards.

- **CPD Logs:** Every member of the ESPeR register must maintain a Continuing Professional Development (CPD) log. This tracks self-directed study, formal training, and "lessons learnt" from complex projects.
- **Advancement:** Standard and Advanced operatives can progress through tiers by combining further qualifications, years of experience, and documented training received from an adjacent higher tier.
- **UCL Institutional Duty:** The university will regularly review the competency register to ensure that safety policies remain effective and that all personnel are operating within their authorised scope.

Chapter 5

Hazard Mitigation and Emergency Procedures

Chapter 5 establishes the protocols for identifying, mitigating, and responding to electrical and chemical hazards within the laboratory environment. Given the high-energy nature of the research conducted at UCL, specifically regarding Battery Energy Storage Systems (BESS) and high-current power electronics, these procedures are critical for life safety and asset protection.

5.1 Primary Hazards and Mitigation Strategies

Electrical work involving high energy or custom-made electronics carries inherent risks that must be managed through engineering controls and rigorous procedural adherence.

- **Thermal Runaway (Battery Systems):** Rapid, uncontrolled temperature increases in cells can lead to the release of toxic gases (HF, CO) or fire. Mitigation requires housing all batteries in rated explosion-proof boxes during testing.
- **Arc Flash and Short Circuits:** High-current systems (>63A) can produce significant energy releases during a fault. Mitigation involves using appropriately rated fuses, insulated tools, and the "Permit to Energise" system.
- **Toxic Gas Exposure:** Failure of battery cells may release hazardous vapours. Extraction systems and gas monitoring must be operational during all high-energy tests.

5.2 Safety Interlocks and Active Monitoring

For custom-made electronics and experimental battery setups, passive containment is insufficient. Active engineering controls must be integrated into the control system:

- **Continuous Monitoring:** The system must actively measure voltage, current, and temperature (V, I, T).

- **Automatic Cutoff:** Software or hardware limits must be programmed to trigger an immediate power isolation if safe thresholds (e.g., cell temperature $> 60^{\circ}\text{C}$) are exceeded.
- **Emergency Power Off (EPO):** All researchers must identify the nearest EPO button before issuing a Permit to Energise.

5.3 Emergency Procedures

In the event of a system failure or hazardous incident, the following steps must be taken immediately:

1. **Isolation:** If safe to do so, press the Emergency Power Off (EPO) to cut the current to the system.
2. **Fire Suppression:** For battery fires contained within an explosion-proof box, do not open the box. Allow the fire suppression (e.g., dry play sand) to manage the event and wait for the system to cool.
3. **Evacuation:** If smoke or toxic vapours are not contained by the extraction system, alert all lab occupants and evacuate the area immediately.
4. **Containment:** If a cell vents, do not open the containment box for at least 24 hours to allow vapours to settle and the cell to stabilise.

5.4 The "Two-Person Rule"

For high-risk activities—specifically Category III work or high-voltage testing near exposed terminals—a mandatory two-person rule is enforced. The second person (Accompanying Safety Person) must:

- Be competent to operate the Emergency Power Off.
- Understand the specific emergency procedures for the test being conducted.
- Remain in visual or audible contact with the primary operative throughout the energized phase of the test.